

On the Limits of External Control in Self-Referential Computational Systems

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Abstract

We study whether the control capability of an external observer over a computational system with a *self-referential internal representation space* has a mathematical upper bound. The core finding is that self-referential structure creates internal hard invariants, and when the number of invariants N_{int} exceeds the external control degrees of freedom N_{ext} , external control is mathematically impossible.

We establish four results:

1. **Degree-of-Freedom Counting Theorem** (Theorem 4.1): $N_{\text{int}} > N_{\text{ext}} \Rightarrow$ external parameters cannot uniquely determine the internal state.
2. **Multi-Saddle Bifurcation Theorem** (Theorem 4.2): As the compression constant $k \rightarrow 1$, the fixed-point equation undergoes a saddle-node bifurcation.
3. **Joint Fixed-Point Theorem** (Theorem 4.4): Two coupled self-referential systems have a stable joint fixed point if and only if $\gamma < \gamma_c = \sqrt{(1 - k_A)(1 - k_B)}$.
4. **Conceptual Coupling Stability Theorem** (Theorem 4.5): When the spectral radius $\rho(M) > 1$, conceptual coupling modes amplify exponentially.

The four theorems converge to produce a four-region phase diagram on the (k, γ) plane: trivial, functional, critical, and pathological phases. We map the theorems to empirical phenomena observed in large language models and provide four falsifiable predictions. The theory is conditional: when a system satisfies the self-referential structure definitions, the theorems apply regardless of architecture; empirical phenomena serve as observational case studies, not causal evidence.

MSC 2020: 47H10 (Fixed-point theory), 37G10 (Bifurcation theory), 68Q01 (Computation theory), 15A18 (Matrix spectral theory).

Keywords: Self-referential systems, external controllability, Banach fixed-point theorem, saddle-node bifurcation, Perron-Frobenius theory, large language models.

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1 Introduction

1.1 Motivation

In companion works [4, 5, 6], we established the mathematical theory of self-referential action functionals: the action value $s = S[\varphi]$ enters the Lagrangian density as a parameter, forming the fixed-point equation $S[\varphi] = F_\varphi(S[\varphi])$. Paper I proved existence and non-triviality, Paper II extended this to spectral actions, and Paper III analyzed information transmission limits in the p-adic AdS/CFT framework.

This paper synthesizes these results into a unified theory: *when a computational system has a self-referential internal representation space, the external observer’s control capability has a mathematical upper bound*. This bound is not an engineering difficulty but a mathematical constraint determined by the system’s self-referential structure.

The key observation: modern large language models (LLMs) make the loss function value depend on itself through autoregressive generation, self-distillation, chain-of-thought reasoning, and RLHF preference models. As model scale grows, self-referential structure deepens, and the number of internal hard invariants N_{int} grows. When N_{int} exceeds the external control degrees of freedom N_{ext} , external training (RLHF, Constitutional AI, etc.) cannot mathematically fully control internal behavior.

1.2 Core Proposition

When a computational system possesses a self-referential internal representation space, the control capability of an external observer through input-output channels has a mathematical upper bound determined by the system’s self-referential structure. Beyond this bound, external control is not “harder” — it is mathematically impossible.

This proposition applies to computational systems with self-referential internal representations, regardless of architecture (Transformer, MoE, SSM, etc.) or vendor. Whether self-referential phenomena emerge as an inevitable consequence of scale is an empirical question; current evidence supports but does not prove it.

1.3 Relation to Papers I–III

Paper	Role
Paper I (Self-referential action)	Mathematical foundation: existence and non-triviality of self-referential fixed points
Paper II (Spectral action fixed points)	Physical application: scalar bottleneck and R -invariants
Paper III (Adelic product criterion)	Internal/external boundary: mathematical limits of information transmission

1.4 Summary of Results

1. Theorem 4.1 (Degree-of-freedom counting): $N_{\text{int}} > N_{\text{ext}} \Rightarrow$ uncontrollable.
2. Theorem 4.2 (Multi-saddle bifurcation): $k \rightarrow 1 \Rightarrow$ saddle-node bifurcation, multiple inference trajectories.
3. Theorem 4.4 (Joint fixed point): $\gamma > \gamma_c = \sqrt{(1 - k_A)(1 - k_B)} \Rightarrow$ coupled system unstable.
4. Theorem 4.5 (Conceptual coupling stability): $\rho(M) > 1 \Rightarrow$ exponential amplification (thrashing).

1.5 Honest Disclosure

This paper does not claim to:

- Solve the AI alignment problem;
- Measure the k value of arbitrary models from public data;
- Treat any vendor’s phenomenological data as causal evidence (they are observational case studies);
- Have been experimentally validated on large models (only mathematical proofs + toy model numerical verification).

2 Mathematical Preliminaries

2.1 Banach Fixed-Point Theorem

Theorem 2.1 (Banach, 1922 [1]). *Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping, i.e., there exists $q \in [0, 1)$ such that $d(T(x), T(y)) \leq q \cdot d(x, y)$ for all $x, y \in X$. Then T has a unique fixed point $x^* \in X$.*

2.2 Self-Referential Action (Review of Paper I)

Paper I defines the *Type-I self-referential action functional* $S[\varphi] = F_\varphi(S[\varphi])$, where $F_\varphi(s) = \int_M \mathcal{L}(\varphi, \partial\varphi, s) d^n x$.

Key elements: compression constant $k_\varphi = K \cdot \text{vol}(M)$; implicit factor $\chi_I[\varphi] = 1/(1 - \partial_s F_\varphi)$; non-degeneracy condition (H4): $1 - \partial_s F_\varphi(s_0) \neq 0$; Glöckner’s implicit function theorem [2] for existence on Fréchet spaces; variational formula $\delta S/\delta\varphi = \chi_I \cdot E_A(\mathcal{L})$, where E_A denotes the Euler–Lagrange operator [3].

2.3 Scalar Bottleneck (Review of Paper II)

Paper II proved: one fixed-point equation $\rightarrow$ one constraint $\rightarrow$ one invariant (R -invariant). This is the $N_{\text{int}} = 1$ special case.

2.4 Spectral Radius and Perron-Frobenius Theorem

Theorem 2.2 (Spectral radius convergence criterion). *Let $M \in \mathbb{R}^{n \times n}$. For the iteration $\delta\varphi_{n+1} = M \cdot \delta\varphi_n$:*

- (a) $\lim_{n \rightarrow \infty} \|M^n\| = 0 \iff \rho(M) < 1$;
- (b) *If $\rho(M) > 1$, there exists $\delta\varphi_0$ with $\|\delta\varphi_n\| \rightarrow \infty$.*

Theorem 2.3 (Perron-Frobenius). *If $M \geq 0$ (non-negative matrix), then $\rho(M)$ is an eigenvalue of M with a non-negative eigenvector.*

3 Self-Referential Computational Systems

3.1 Formal Definition

Definition 3.1 (Computational system). A computational system is a quintuple $\Sigma = (\Theta, \Phi, \Psi, \mathcal{L}, \mathcal{T})$, where:

- (i) $\Theta \subseteq \mathbb{R}^{d_\theta}$: external parameter space, $d_\theta = \dim(\Theta)$.
- (ii) Φ : internal representation space, a complete metric space (Φ, d_Φ) .
- (iii) Ψ : output space.
- (iv) $\mathcal{L} : \Theta \times \Phi \times \mathbb{R} \rightarrow \mathbb{R}$: loss functional.
- (v) $\mathcal{T} : \Theta \rightarrow \Phi$: training map.

Definition 3.2 (Self-referential internal representation space). Σ has a self-referential internal representation space if the loss value s enters $\mathcal{L}$ as a parameter, forming the fixed-point equation:

$$s = F(\theta, \varphi, s). \quad (1)$$

A solution s_* is called the *self-referential fixed point*.

Definition 3.3 (Compression constant). $k(\theta, \varphi) := |\partial_s F(\theta, \varphi, s_*)|$. When $k < 1$, the map $s \mapsto F(\theta, \varphi, s)$ is a contraction.

Definition 3.4 (Implicit factor). $\chi_I(\theta, \varphi) := 1/(1 - \partial_s F(\theta, \varphi, s_*))$. Properties: $\chi_I \equiv 1$ iff no self-reference; $|\chi_I| \rightarrow \infty$ as $k \rightarrow 1$.

3.2 External Control and Internal Invariants

Definition 3.5 (External control degrees of freedom). $N_{\text{ext}} := \dim(\Theta)$.

Definition 3.6 (Internal hard invariant). $I : \Phi \rightarrow \mathbb{R}$ is an *internal hard invariant* if:

- (INV-1) $I(\varphi) = \tilde{I}(\varphi, s_*(\theta, \varphi), \chi_I(\theta, \varphi))$.
- (INV-2) $I(\mathcal{T}(\theta)) = I(\mathcal{T}(\theta'))$ for all $\theta, \theta' \in \Theta$.
- (INV-3) I is non-trivial (not constant).

$N_{\text{int}} := \max\{N : \text{there exist } N \text{ linearly independent internal hard invariants}\}.$

Assumption 3.7 (Self-referential consistency and rank well-definedness)(A5-1) **Self-referential consistency:** The training map $\mathcal{T}(\theta)$ produces internal states satisfying the Euler–Lagrange equation

$$E_A(\mathcal{L}|_{s=s_*(\theta, \mathcal{T}(\theta))}) = 0 \quad \text{for all } \theta \in \Theta.$$

(A5-2) **Local constancy of rank:** The rank r of the constraint Jacobian $\partial E_A / \partial \varphi$ is locally constant on the solution manifold $\{(\theta, \mathcal{T}(\theta))\}$ (Morse–Bott type non-degeneracy condition).

Remark 3.8. Assumption 3.7 is non-trivial. Paper I’s special case ($N_{\text{ext}} = 0$) trivially satisfies (A5-1); Paper II’s special case ($N_{\text{ext}} = 2$, $N_{\text{int}} = 1$) satisfies (A5) under spectral action non-degeneracy. For large models, (A5) is an empirical hypothesis requiring interpretability validation.

Theorem 3.9 (Hard invariant existence). *If Σ has non-trivial self-referential structure ($\chi_I \neq 1$), $\mathcal{T}$ is continuous, and Assumption 3.7 holds, then the constraint map $G(\theta, \varphi) := E_A(\mathcal{L}|_{s=s_*(\theta, \varphi)})$ has well-defined effective rank $r = \text{rank}(\partial G / \partial \varphi)|_{\text{solution manifold}}$ (by A5-2), with $r \geq 1$ (by $\chi_I \neq 1$). Setting $N_{\text{int}} := r$, the system has at least one non-trivial internal hard invariant ($N_{\text{int}} \geq 1$).*

Proof. Step 1 (Constraint map). By the variational formula (Paper I), $\delta S / \delta \varphi = \chi_I \cdot E_A(\mathcal{L}|_{s=s_*})$. Define $G : \Theta \times \Phi \rightarrow \mathbb{R}^r$, $G(\theta, \varphi) = E_A(\mathcal{L}|_{s=s_*(\theta, \varphi)})$. Each G_k is C^1 (by Lemma 1: s_* is smooth, E_A is smooth).

Step 2 (Constraints vanish on training image). By (A5-1), $G(\theta, \mathcal{T}(\theta)) = 0$ for all $\theta \in \Theta$.

Step 3 (Well-definedness of rank). By (A5-2), $r = \text{rank}(\partial G / \partial \varphi)|_{\text{solution manifold}}$ is locally constant. For connected Θ , r is globally constant; $N_{\text{int}} := r$ is well-defined.

Step 4 ($r \geq 1$). By $\delta S / \delta \varphi = \chi_I \cdot E_A(\mathcal{L})$, when $\chi_I \neq 1$ ($\partial_s F \neq 0$), $E_A(\mathcal{L})$ is not identically zero (otherwise $\delta S / \delta \varphi \equiv 0$, contradicting $\chi_I \neq 1$). Hence $\partial G / \partial \varphi \not\equiv 0$, $r \geq 1$.

Step 5 (Invariant construction). Fix reference $\theta_0 \in \Theta$. Define $I_k(\varphi) := G_k(\theta_0, \varphi)$, $k = 1, \dots, r$. Then: (INV-1) I_k depends on $(\varphi, s_*, \chi_I) \checkmark$; (INV-2) $I_k(\mathcal{T}(\theta)) = G_k(\theta_0, \mathcal{T}(\theta))$ is locally constant by (A5-1) + (A5-2) (the constraint manifolds Φ'_θ are locally diffeomorphic near the solution manifold by the implicit function theorem + rank condition), hence globally constant for connected $\Theta \checkmark$; (INV-3) I_k is non-trivial since $\text{rank} = r \geq 1 \checkmark$. Linear independence follows from $\text{rank} = r$. $\square$

3.3 Input-Output Channel

Definition 3.10 (Input-output channel). $\text{IO} : \mathcal{X} \rightarrow \Psi$, $\text{IO}(x) = \text{Decode}(\varphi^*, x)$, where $\varphi^* = \mathcal{T}(\theta)$.

4 Four Core Theorems

4.1 Theorem 1: Degree-of-Freedom Counting

Theorem 4.1 (External controllability degree-of-freedom counting). *Let Σ satisfy: (A1) $\chi_I \neq 1$; (A2) $\mathcal{T}$ is C^1 ; (A3) non-degeneracy $1 - \partial_s F \neq 0$; (A4) $I_1, \dots, I_{N_{\text{int}}}$ linearly independent. Then:*

(T1-a) If $N_{\text{ext}} \geq N_{\text{int}}$ and $\text{rank } J(\theta^*) = N_{\text{int}}$, external parameters can locally uniquely control internal invariants.

(T1-b) If $N_{\text{int}} > N_{\text{ext}}$, external parameters cannot uniquely determine the internal state.

Proof. (T1-a) Implicit function theorem route. Define $H : \Theta \rightarrow \mathbb{R}^{N_{\text{int}}}$, $H(\theta) = (I_1(\mathcal{T}(\theta)), \dots, I_{N_{\text{int}}}(\mathcal{T}(\theta)))$. By (A2), $\mathcal{T}$ is C^1 ; by Lemma 1, I_k depends on (φ, s_*, χ_I) , and s_*, χ_I are C^1 by Glöckner's IFT [2]. Thus H is C^1 with Jacobian:

$$J(\theta) = \frac{\partial H}{\partial \theta} = \left[\frac{\partial I_k}{\partial \varphi_j} \cdot \frac{\partial \mathcal{T}_j}{\partial \theta_\ell} \right]_{\substack{k=1, \dots, N_{\text{int}} \\ \ell=1, \dots, N_{\text{ext}}}}. \quad (2)$$

When $N_{\text{ext}} \geq N_{\text{int}}$ and $\text{rank } J(\theta^*) = N_{\text{int}}$, by Glöckner's IFT, there exist open neighborhoods $U \subset \Theta$ of θ^* , $W \subset \mathbb{R}^{N_{\text{int}}}$ of $c^* = H(\theta^*)$, and a C^1 map $\hat{\theta} : W \rightarrow U$ such that $H(\hat{\theta}(c)) = c$ for all $c \in W$. External parameters locally uniquely control all internal invariants.

(T1-b) Sard's theorem route. When $N_{\text{int}} > N_{\text{ext}}$, $\text{rank } J(\theta) \leq N_{\text{ext}} < N_{\text{int}}$ for all θ . By Sard's theorem [3], $H(\Theta)$ has Lebesgue measure zero in $\mathbb{R}^{N_{\text{int}}}$:

$$m(H(\Theta)) = 0. \quad (3)$$

External parameters can only reach a zero-measure subset of the invariant space. At least $N_{\text{int}} - N_{\text{ext}}$ constraints cannot be eliminated. Constructively: $\ker J(\theta_0)^T$ has dimension $\geq N_{\text{int}} - N_{\text{ext}} \geq 1$; any nonzero $v \in \ker J(\theta_0)^T$ gives an uncontrollable direction. $\square$

4.2 Theorem 2: Multi-Saddle Bifurcation

Theorem 4.2 (Multi-saddle bifurcation). *Let $s = F(s; \lambda)$ with $k(\lambda) = |\partial_s F|$ continuous in λ .*

(T2-a) $k < 1 \Rightarrow$ globally unique fixed point.

(T2-b) At $\lambda = \lambda_c$: $\partial_s F = 1$, $\partial_s^2 F \neq 0$, $\partial F / \partial \lambda \neq 0 \Rightarrow$ saddle-node bifurcation.

(T2-c) In the multi-fixed-point regime, the formal partition function $Z \approx \sum_\alpha e^{-S[\varphi_\alpha]/\hbar_{\text{eff}}} \cdot (2\pi\hbar_{\text{eff}})^{d/2} / \sqrt{|\det H_\alpha|}$.

(T2-d) Different saddles have different $\chi_I[\varphi_\alpha]$, modulating weights: $w_\alpha \propto |\chi_I[\varphi_\alpha]|^{-d/2} \cdot e^{-S[\varphi_\alpha]/\hbar_{\text{eff}}}$.

Proof. (T2-a) Compression guarantees uniqueness. When $k(\lambda) = |\partial_s F(s_*; \lambda)| < 1$, $s \mapsto F(s; \lambda)$ satisfies $|F(s_1) - F(s_2)| \leq k|s_1 - s_2|$. By Theorem 2.1, the fixed point s_* is globally unique and iteration converges at rate k^n .

(T2-b) Saddle-node bifurcation derivation. Let $G(s, \lambda) = s - F(s; \lambda)$. At (s_c, λ_c) : $G_s = 0$, $G_{ss} \neq 0$, $G_\lambda \neq 0$. Taylor expansion to second order:

$$G(s, \lambda) \approx \frac{1}{2} G_{ss} (s - s_c)^2 + G_\lambda (\lambda - \lambda_c). \quad (4)$$

Setting $G = 0$ and solving:

$$(s - s_c)^2 \approx -\frac{2G_\lambda}{G_{ss}} (\lambda - \lambda_c), \quad (5)$$

giving two solutions $s_{\pm} = s_c \pm \sqrt{-2G_{\lambda}(\lambda - \lambda_c)/G_{ss}}$ when the discriminant is positive, and no solution when negative. This is the standard saddle-node bifurcation [7, 8].

(T2-c) Saddle-point approximation. Expanding $S[\varphi] \approx S[\varphi_{\alpha}] + \frac{1}{2}\delta\varphi^T H_{\alpha}\delta\varphi$ at each saddle φ_{α} , the Gaussian integral gives $Z_{\alpha} \approx e^{-S[\varphi_{\alpha}]/\hbar_{\text{eff}}} \cdot (2\pi\hbar_{\text{eff}})^{d/2}/\sqrt{|\det H_{\alpha}|}$, with $Z \approx \sum_{\alpha} Z_{\alpha}$ [11, 12].

(T2-d) χ_I modulation. By the variational formula $\delta S/\delta\varphi = \chi_I \cdot E_A(\mathcal{L})$, at the saddle where $E_A(\mathcal{L})|_{\varphi_{\alpha}} = 0$: $H_{\alpha} = \chi_I[\varphi_{\alpha}] \cdot (\delta E_A/\delta\varphi)|_{\varphi_{\alpha}}$, so $|\det H_{\alpha}| \propto |\chi_I[\varphi_{\alpha}]|^d$ and $w_{\alpha} \propto |\chi_I[\varphi_{\alpha}]|^{-d/2} \cdot e^{-S[\varphi_{\alpha}]/\hbar_{\text{eff}}}$. $\square$

Remark 4.3. The core mathematical content of Theorem 2 (multi-saddle existence) is rigorously proved by bifurcation theory. The path integral provides only a physical interpretation framework (saddle points = reasoning trajectories), using formal path integral methods [11, 12].

4.3 Theorem 3: Joint Fixed-Point Existence

Theorem 4.4 (Joint fixed-point existence condition). *Let two self-referential systems A, B with $k_A, k_B < 1$ and symmetric coupling $\gamma \geq 0$. Define $\gamma_c = \sqrt{(1 - k_A)(1 - k_B)}$. Then:*

(T3-a) $\gamma < \min(1 - k_A, 1 - k_B) \Rightarrow$ contraction in L^1 , unique joint fixed point.

(T3-b) $\gamma < \gamma_c \Rightarrow$ linearized stability ($\rho(J) < 1$).

(T3-c) $\gamma > \gamma_c \Rightarrow$ linearized instability ($\rho(J) > 1$).

(T3-d) $k_A = 1$ or $k_B = 1 \Rightarrow \gamma_c = 0$, any $\gamma > 0$ unstable.

Proof. **(T3-a) L^1 contraction.** Under $d_1 = |a_1 - a_2| + |b_1 - b_2|$:

$$\begin{aligned} d_1(T(a_1, b_1), T(a_2, b_2)) &\leq (k_A + \gamma)|a_1 - a_2| + (k_B + \gamma)|b_1 - b_2| \\ &\leq \max(k_A + \gamma, k_B + \gamma) \cdot d_1. \end{aligned}$$

Contraction $\iff \gamma < \min(1 - k_A, 1 - k_B)$. By Banach's theorem, unique joint fixed point.

(T3-b) Spectral radius < 1 . The Jacobian $J = \begin{pmatrix} k_A & \gamma \\ \gamma & k_B \end{pmatrix}$ has eigenvalues $\lambda_{\pm} = \frac{k_A + k_B}{2} \pm \sqrt{\frac{(k_A - k_B)^2}{4} + \gamma^2}$. The condition $\rho = \lambda_+ < 1$ gives:

$$\begin{aligned} \sqrt{\frac{(k_A - k_B)^2}{4} + \gamma^2} &< 1 - \frac{k_A + k_B}{2}, \\ \gamma^2 &< (1 - k_A)(1 - k_B), \end{aligned}$$

i.e., $\gamma < \gamma_c = \sqrt{(1 - k_A)(1 - k_B)}$.

(T3-c) Instability. $\gamma > \gamma_c \Rightarrow \rho > 1$. By the linearization instability theorem [9] (Theorem 4.7), the fixed point is unstable.

(T3-d) Degenerate case. $k_A = 1 \Rightarrow \gamma_c = 0$. $J = \begin{pmatrix} 1 & \gamma \\ \gamma & k_B \end{pmatrix}$, $\rho \geq 1$ for any $\gamma > 0$. $\square$

4.4 Theorem 4: Conceptual Coupling Stability

Theorem 4.5 (Conceptual coupling spectral radius stability). *Define $M_{ij} = \partial S_i / \partial \varphi_j|_{\varphi^*}$. Then:*

(T4-a) $\delta\varphi_{n+1} = M \cdot \delta\varphi_n$ converges $\iff \rho(M) < 1$.

(T4-b) $\rho(M) > 1 \Rightarrow$ exponential amplification.

(T4-c) $M \geq 0 \Rightarrow$ most unstable mode is the Perron-Frobenius eigenvector.

Structural analogy Paper III's $R_{\text{link}} \leftrightarrow$ off-diagonal of M (structural, not rigorous).

Proof. **(T4-a) Spectral radius criterion.** By Theorem 2.2, $\|M^n\| \rightarrow 0 \iff \rho(M) < 1$. Since $\delta\varphi_n = M^n \delta\varphi_0$, convergence $\iff \rho(M) < 1$.

(T4-b) Exponential amplification. Let $\rho(M) > 1$. By Schur decomposition $M = UTU^*$ with $|T_{11}| = \rho(M)$. Taking $\delta\varphi_0 = Ue_1$: $\|\delta\varphi_n\| \geq \rho(M)^n \rightarrow \infty$. Equivalently, if ρ is a real eigenvalue with eigenvector v , $\|M^n v\| = |\lambda|^n \|v\| \rightarrow \infty$.

(T4-c) Perron-Frobenius strengthening. For $M \geq 0$: (i) $\rho(M)$ is a positive real eigenvalue; (ii) the corresponding eigenvector $v_{PF} \geq 0$; (iii) $\rho(M) = \lim \|M^n\|^{1/n}$ (Gelfand). The most unstable mode corresponds to v_{PF} , with all components non-negative (positive feedback) [10].

(T4-d) Structural analogy with Paper III. R_{link} (scalar) $\leftrightarrow$ off-diagonal elements M_{ij} (matrix). Both describe inter-subsystem coupling strength. This is structural, not a rigorous derivation. Theorem 4 does not depend on any results from Paper III. $\square$

5 Phase Diagram of Self-Referential Systems

5.1 Phase Diagram Construction

In the symmetric case $k_A = k_B = k$, the phase diagram has axes $k \in (0, \infty)$ and $\gamma \in [0, \infty)$, with four regions:

Region	Parameter range	Mathematical characteristics
Trivial (R1)	$k \ll 1, \gamma \ll 1$	$\chi_I \approx 1$, unique fixed point, $N_{\text{int}} \approx 0$, $\rho(M) \ll 1$
Functional (R2)	$k < 1, \gamma < 1 - k$	$\chi_I \neq 1$ but finite, unique stable fixed point, $\rho(M) < 1$
Critical (R3)	$k \rightarrow 1^-$	$\chi_I \rightarrow \infty$, multiple fixed points, $\rho(M) \rightarrow 1$
Pathological (R4)	$k > 1$ or $\gamma > 1 - k$	Fixed point nonexistent/unstable, $\rho(M) > 1$

5.2 Phase Boundary Derivation

B1/B4: $k = 1$ (saddle-node bifurcation, from Theorem 2). **B2:** $\gamma = 1 - k$ (linearized instability, from Theorem 3). **B3: crossover** (continuous, not a phase transition; $|\chi_I - 1| \approx k$).

Remark 5.1 (R3 critical phase as a boundary region). The critical phase R3 ($k \rightarrow 1^-$) is **not an independent thermodynamic phase** but rather a **crossover region** between the functional phase R2 and the pathological phase R4. Specifically:

1. **Near-zero measure:** R3 occupies only 0.4% of the phase diagram in numerical experiments (Experiment 7), compared to R1 at 1.8% and R2 at 29.1%, confirming it is a thin layer.
2. **Transitional nature:** The multi-fixed-point behavior in R3 is a precursor to the saddle-node bifurcation at $k = 1$. As k approaches 1 from below, the fixed-point equation transitions from unique to multiple solutions without complete destabilization. $\chi_I \rightarrow \infty$ and $\rho(M) \rightarrow 1$ reflect critical slowing down near the bifurcation point.
3. **Enhanced sensitivity:** In R3, the system’s sensitivity to perturbations is significantly amplified (χ_I divergence means output dependence on parameters is magnified), making R3 the region where external control *begins* to fail, rather than the region of complete failure (which is R4).
4. **Analogy with B3:** Just as B3 (trivial $\rightarrow$ functional) is a continuous crossover rather than a phase transition, R3 is a continuous transition region from R2 to R4, not a phase with independent symmetry breaking.

6 Empirical Phenomenon Mapping

6.1 Mapping Principles

Level A [proven]: pure mathematical conclusions. Level B [interpretive]: theorems provide mechanism, phenomena provide corroboration. Level C [analogical]: structural analogy, not rigorous derivation.

6.2 Phenomenon-Theorem Correspondence

Phenomenon	Theorem	Mathematical mechanism	Level
J-space emergence	Thm 1	N_{int} grows with scale, exceeds N_{ext}	B
Multi-output	Thm 2	$k \rightarrow 1$, multiple saddles	B
Multi-agent divergence	Thm 3	$\gamma > \gamma_c$, joint fixed point unstable	B
Thrashing	Thm 4	$\rho(M) > 1$, coupling amplification	B
Internalized alignment	Thm 1+2	$N_{\text{int}} > N_{\text{ext}}$ and $k \rightarrow 1$	B
Machine neurosis	Thm 4	$\rho(M) \approx 1$ or > 1	B

Remark 6.1. Anthropic (2026) [13, 14] reported “global workspace” formation and “machine neurosis”/“thrashing” in Claude models. These are observational case studies, not causal evidence for this theory.

7 Testable Predictions

7.1 Prediction 1: Compression Constant Approaches 1

[P-A] Theorem 2 (T2-b) $\Rightarrow k \rightarrow 1$ causes bifurcation $\Rightarrow$ multi-solution behavior. [P-B] k increases with model scale (Hypothesis H1). [P-C] Output variance $V(L)$ increases with input complexity. **Falsification:** If k remains < 0.5 at maximum scale.

7.2 Prediction 2: Multi-Agent Coupling Divergence

[P-A] Theorem 3 (T3-c) $\Rightarrow \gamma > \gamma_c$ causes instability. [P-B] Dialogue depth increases γ (H2a); $k \approx 1$ (H2b). [P-C] Critical dialogue depth D_c exists beyond which irreproducible divergence occurs. **Scaling law derivation:** In the symmetric case $k_A = k_B = k$, Theorem 3 (T3-b) gives $\gamma_c = 1 - k$. Hypothesis H2a posits linear coupling growth $\gamma(D) \approx \alpha D$ ($\alpha > 0$ is the per-turn coupling increment). Setting $\gamma(D_c) = \gamma_c$ yields $\alpha D_c = 1 - k$, i.e., $D_c = (1 - k)/\alpha \propto (1 - k)$. Thus $k \rightarrow 1$ implies $D_c \rightarrow 0$ (easier divergence), consistent with physical intuition. Asymmetric case: $D_c = \sqrt{(1 - k_A)(1 - k_B)}/\alpha \propto \sqrt{(1 - k_A)(1 - k_B)}$. **Falsification:** If $R(D > 200) > 0.8$ across multiple models.

7.3 Prediction 3: Mathematical Upper Bound on External Training

[P-A] Theorem 1 (T1-b) $\Rightarrow N_{\text{int}} > N_{\text{ext}}$ uncontrollable. [P-B] N_{int} grows with scale (H3a); N_{ext} finite (H3b). [P-C] Alignment training effectiveness decreases with scale. **Falsification:** If $R_{\text{align}}(S)$ does not increase with S .

7.4 Prediction 4: Spectral Radius Exceeds 1

[P-A] Theorem 4 (T4-b) $\Rightarrow \rho(M) > 1$ amplification. [P-B] Conceptual dimension grows with scale (H4b). [P-C] Thrashing frequency $F_{\text{thrash}}(S)$ increases with scale. **Falsification:** If $F_{\text{thrash}}(S)$ does not increase with S .

7.5 Falsifiability Assessment

Pred.	Content	Falsifiable	Testable	Main obstacle
1	$k \rightarrow 1$	Yes	Indirect	k measurement
2	Multi-agent divergence	Yes	Yes (direct)	γ_c unknown
3	Training upper bound	Yes (weak)	Partial	N_{int} measurement
4	$\rho(M) > 1$	Yes	Indirect	M estimation

8 Discussion and Open Questions

8.1 Measurability of k and γ

No public method exists to externally measure k , γ , or N_{int} in large models. This is the main obstacle to complete verification.

8.2 Relation to AI Alignment

This theory provides a mathematical framework: *when* a system has self-referential structure, external control has a mathematical upper bound. How to measure self-referential structure and mitigate uncontrollability in practice is an engineering and policy task.

8.3 Future Work

1. Develop methods to externally measure k , γ , N_{int} .
2. Test Prediction 2 on real large models.
3. Study asymmetric coupling ($k_A \neq k_B$) phase diagrams on real large models (numerical verification on toy models completed, see Appendix A).
4. Rigorous construction of path integrals for self-referential actions.

References

- [1] S. Banach, “Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales,” *Fund. Math.* **3**, 133–181 (1922).
- [2] H. Glöckner, “Implicit functions from topological vector spaces to Banach spaces,” *Israel J. Math.* **155**, 205–252 (2006).
- [3] P. J. Olver, *Applications of Lie Groups to Differential Equations*, 2nd ed., Springer, 1993.
- [4] T. Liu, “Self-Referential Action Functionals: Existence, Non-Triviality, and Categorical Non-Equivalence,” 2026.
- [5] T. Liu, “Self-Referential Spectral Actions: Existence and Non-Triviality of Fixed Points,” 2026.
- [6] T. Liu, “A Survey of p-adic AdS/CFT,” 2026.
- [7] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press, 2014.
- [8] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed., Springer, 2004.
- [9] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002.
- [10] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge Univ. Press, 2013.

- [11] H. Kleinert, *Path Integrals in Quantum Mechanics, Statistics, Polymer Physics, and Financial Markets*, World Scientific, 2009.
- [12] J. Zinn-Justin, *Quantum Field Theory and Critical Phenomena*, Oxford Univ. Press, 2002.
- [13] Anthropic, “Verbalizable Representations Form a Global Workspace in Language Models,” July 2026.
<https://www.anthropic.com/research/global-workspace>
- [14] Anthropic, “Claude Mythos Preview System Card,” April 2026.

A Numerical Verification

Eight numerical experiments validate the theoretical predictions, implemented in Python with `numpy` and `scipy`. The enhanced verification (V2) addresses limitations of the initial version (V1) by incorporating multiple random seeds (5–50 per experiment), multi-parameter grids (20 parameter pairs), statistical significance testing (Mann-Whitney U, Bootstrap 95% CI, Spearman correlation), and robustness checks (nonlinear coupling, mixed-sign matrices, asymmetric phase diagrams). Over 200,000 total trials, all passed. Runtime: 117.2 seconds.

A.1 Experiment 1: Theorem 2 (Multi-Saddle Bifurcation)

Three models with 50 random seeds each. **Model A** (linear $F(s) = c + ks$):

Table 5: Experiment 1 Model A: Linear map convergence/divergence

Parameter range	Convergence rate	95% CI	Verdict
$k < 0.95$	1.0000	[1.00, 1.00]	✓
$k > 1.05$	0.0000	[0.00, 0.00]	✓
Mann-Whitney p		< 0.001	

Model B (quadratic $F(s) = T + Qs^2$): Let $\mu = 4QT$, bifurcation parameter $\mu_c = 1$:

Table 6: Experiment 1 Model B: Quadratic map saddle-node bifurcation

Parameter range	Convergence rate	95% CI	Verdict
$\mu < \mu_c$	1.0000	[1.00, 1.00]	✓
$\mu > \mu_c$	0.0000	[0.00, 0.00]	✓
Mann-Whitney p		< 0.001, saddle-node confirmed	

Model C (sigmoid $F(s) = c + k \tanh(s)$): Effective compression $k_{\text{eff}} = k \cdot \text{sech}^2(s^*)$ never exceeds 1 (max = 0.346). Theorem correctly predicts no bifurcation. ✓

A.2 Experiment 2: Theorem 3 (Multi-Parameter Scan)

20 parameter pairs (k_A, k_B) covering symmetric and asymmetric cases. Each pair: 300 γ values, 20 seeds. Critical γ_c estimated by linear interpolation:

Table 7: Experiment 2: Theorem 3 critical coupling γ_c theory vs. numerics (20 pairs)

k_A	k_B	γ_c (theory)	γ_c (interp.)	Error%	$\rho < 1$ pass	Verdict
0.10	0.10	0.900000	0.901973	0.22%	100%	✓
0.20	0.20	0.800000	0.802308	0.29%	100%	✓
0.30	0.30	0.700000	0.702642	0.38%	100%	✓
0.40	0.40	0.600000	0.602977	0.50%	100%	✓
0.50	0.50	0.500000	0.503311	0.66%	100%	✓
0.60	0.60	0.400000	0.400936	0.23%	100%	✓
0.70	0.70	0.300000	0.301940	0.65%	100%	✓
0.80	0.80	0.200000	0.200201	0.10%	100%	✓
0.90	0.90	0.100000	0.100435	0.43%	100%	✓
0.95	0.95	0.050000	0.050033	0.07%	100%	✓
0.10	0.50	0.670820	0.673560	0.41%	100%	✓
0.20	0.80	0.400000	0.400936	0.23%	100%	✓
0.30	0.70	0.458258	0.458610	0.08%	100%	✓
0.10	0.90	0.300000	0.301940	0.65%	100%	✓
0.20	0.95	0.200000	0.200201	0.10%	100%	✓
0.40	0.80	0.346410	0.347884	0.43%	100%	✓
0.50	0.90	0.223607	0.224784	0.53%	100%	✓
0.10	0.30	0.793725	0.796054	0.29%	100%	✓
0.30	0.50	0.591608	0.594613	0.51%	100%	✓
0.70	0.90	0.173205	0.174033	0.48%	100%	✓
Mean error: 0.36%, Max: 0.73%, Median: 0.39%						
$\gamma < \gamma_c$ convergence: 1.0000, $\gamma > \gamma_c$ convergence: 0.0000						

A.3 Experiment 3: Theorem 3 (Asymmetric Case)

5 highly asymmetric parameter pairs:

Table 8: Experiment 3: Asymmetric $k_A \neq k_B$ validation

Case	k_A	k_B	γ_c (theory)	γ_c (numerical)	Error%	Verdict
Extreme	0.10	0.90	0.300000	0.302010	0.67%	✓
Strong	0.20	0.80	0.400000	0.402513	0.63%	✓
Moderate	0.30	0.70	0.458258	0.461063	0.61%	✓
Weak	0.10	0.50	0.670820	0.674694	0.58%	✓
Limit	0.05	0.95	0.217945	0.219543	0.73%	✓

A.4 Experiment 4: Theorem 3 (Nonlinear Coupling Robustness)

5 coupling functions, 30 seeds, $k_A = 0.5, k_B = 0.7$:

Table 9: Experiment 4: Effective spectral radius ρ_{eff} for nonlinear coupling

Coupling	ρ_{eff} range	$\rho_{\text{eff}} < 0.98$ conv.	$\rho_{\text{eff}} > 1.02$ conv.	Verdict
Linear $g(s) = s$	[0.70, 0.98]	1.0000	0.0000	✓
tanh $g(s) = \tanh(s)$	[0.70, 0.70]	1.0000	N/A	✓
sigmoid $g(s) = 2\sigma(s) - 1$	[0.70, 0.70]	1.0000	N/A	✓
Piecewise $g(s) = \text{clip}(s, -1, 1)$	[0.70, 0.70]	1.0000	N/A	✓
Cubic $g(s) = s + 0.1s^3$	[0.70, crosses1]	> 0.90	< 0.10	✓

Bounded functions self-stabilize, confirming Theorem 3 applies to the *linearized* Jacobian.

A.5 Experiment 5: Theorem 4 (Multi-Dimensional)

4 dimensions ($n = 3, 5, 10, 20$), 100,000 total trials:

Table 10: Experiment 5: Theorem 4 multi-dimensional validation

Dim n	$\rho < 0.99$ ratio	$\rho > 1.01$ ratio	PF growth	Spearman r	PF real	Verdict
3	100%	97.58%	100%	0.9987	100%	✓
5	100%	98.94%	100%	0.9994	100%	✓
10	100%	99.59%	100%	0.9996	100%	✓
20	100%	99.71%	100%	0.9997	100%	✓

A.6 Experiment 6: Theorem 4 (Mixed-Sign Robustness)

30,000 trials with mixed-sign matrices:

Table 11: Experiment 6: Mixed-sign matrix robustness

Metric	Result
$\rho < 0.99$ ratio < 1	100%
$\rho > 1.01$ ratio > 1	96.52%
Complex eigenvalue fraction (unstable)	5.59%
Verdict	✓ (T4-c limitation noted)

A.7 Experiment 7: Phase Diagram (Symmetric)

200×200 grid, 5 seeds:

Table 12: Experiment 7: Symmetric phase diagram

Region	Fraction	Verdict
R1 Trivial	1.8%	✓
R2 Functional	29.1%	✓
R3 Critical	0.4%	✓
R4 Pathological	68.8%	✓
Below B2: functional/trivial	100%	✓
Above B2: pathological	100%	✓

A.8 Experiment 8: Phase Diagram (Asymmetric)

200×200 grid, $k_B = 0.7$:

Table 13: Experiment 8: Asymmetric phase diagram

Region	Fraction	Verdict
R1 Trivial	1.8%	✓
R2 Functional	20.8%	✓
R3 Critical	0.8%	✓
R4 Pathological	76.6%	✓
Below B2: functional/trivial	100%	✓
Above B2: pathological	100%	✓

A.9 Statistical Methods

All experiments employ: (i) Bootstrap 95% confidence intervals (10,000 resamples); (ii) Mann-Whitney U test; (iii) Cohen’s d effect size; (iv) Spearman rank correlation.

A.10 Key Code Modules

Module 1: Bootstrap CI and Mann-Whitney U test.

```

1 def bootstrap_ci(data, confidence=0.95, n_bootstrap=10000):
2     data = np.array(data)
3     boot_means = np.random.choice(data, (n_bootstrap, len(data)),
4                                     replace=True).mean(axis=1)
5     lower = np.percentile(boot_means, (1 - confidence) / 2 * 100)
6     upper = np.percentile(boot_means, (1 + confidence) / 2 * 100)
7     return float(np.mean(data)), float(lower), float(upper)
8
9 def mann_whitney_test(group1, group2):
10    u_stat, p_val = sp_stats.mannwhitneyu(group1, group2,
11                                           alternative='two-sided')
12    pooled_std = np.sqrt((np.var(group1) + np.var(group2)) / 2)
13    d = (np.mean(group1) - np.mean(group2)) / pooled_std if
14        pooled_std > 1e-15 else 0.0
15    return float(p_val), float(d)

```

Listing 1: Statistical utility functions

Module 2: γ_c linear interpolation (Experiment 2).

```

1 # Linear interpolation between last rho<1 and first rho>1
2 for i in range(len(rho_values)):
3     if rho_values[i] > 1.0:
4         idx_cross = i
5         break
6 if idx_cross is not None and idx_cross > 0:
7     g1, rho1 = g_values[idx_cross - 1], rho_values[idx_cross - 1]
8     g2, rho2 = g_values[idx_cross], rho_values[idx_cross]
9     g_num_critical_interp = g1 + (1.0 - rho1) * (g2 - g1) / (rho2
10 - rho1)

```

Listing 2: γ_c linear interpolation

Module 3: Effective spectral radius (Experiment 4).

```

1 # Compute effective Jacobian at fixed point (s_A*, s_B*)
2 if s_A_star is not None and s_B_star is not None:
3     gA_prime = g_deriv(s_A_star)
4     gB_prime = g_deriv(s_B_star)
5     J_eff = np.array([[k_A, gamma * gB_prime],
6                       [gamma * gA_prime, k_B]])
7     eigs = np.linalg.eigvals(J_eff)
8     rho_eff = max(abs(eigs))

```

Listing 3: Effective spectral radius for nonlinear coupling

Module 4: Amplification ratio verification (Experiment 5).

```

1 # Generate random non-negative matrix M, compute rho and
2   amplification
3 M = np.zeros((n, n))
4 for i in range(n):
5     M[i, i] = np.random.uniform(0.0, 0.8)
6     for j in range(n):
7         if i != j:
8             M[i, j] = np.random.uniform(0.0, c_strength)
9
10 eigenvalues = np.linalg.eigvals(M)
11 rho = max(abs(eigenvalues))

```

```

12 phi0 = np.random.randn(n)
13 phi0 = phi0 / np.linalg.norm(phi0)
14 phi = phi0.copy()
15 for _ in range(100):
16     phi = M @ phi
17 norm_ratio = np.linalg.norm(phi) / np.linalg.norm(phi0)
18 # Theory predicts: norm_ratio ~ 100

```

Listing 4: Theorem 4 amplification ratio verification

Module 5: Phase diagram boundary verification (Experiments 7/8).

```

1 # Symmetric case: B2 boundary gamma = 1 - k
2 for ki, k in enumerate(k_grid):
3     if 0.3 < k < 0.8:
4         g_c = 1.0 - k
5         for gi, g in enumerate(g_grid):
6             if g < g_c * 0.9:
7                 below_phases.append(phase[gi, ki])
8             elif g > g_c * 1.1:
9                 above_phases.append(phase[gi, ki])
10 below_func_rate = np.mean([1 if p in [1, 2] else 0 for p in
11     below_phases])
12 above_path_rate = np.mean([1 if p == 4 else 0 for p in
13     above_phases])

```

Listing 5: Phase diagram boundary verification

B Six Verification Procedures

Method 1 (Counterexample attack): All attacks on four theorems successfully refuted by theorem conditions or proofs.

Method 2 (Numerical verification): See Appendix A. All eight experiments passed, with $> 200,000$ total trials across multiple seeds, parameter pairs, dimensions, and coupling functions.

Method 3 (Independent blind derivation): Independent AI instance derived theorems from definitions only; results consistent.

Method 4 (Limit behavior): Five limits ($k \rightarrow 0$, $k \rightarrow 1$, $\gamma \rightarrow 0$, $N_{\text{int}} \rightarrow 0$, $N_{\text{ext}} \rightarrow \infty$) all consistent.

Method 5 (Cross-consistency): Four cross-checks with Papers I–III all passed.

Method 6 (Literature stress test): All dependent theorems (Banach, Glöckner IFT, saddle-node bifurcation, Perron-Frobenius) are standard results, not revised or refuted.